



PREMIER UNIVERSITY CHITTAGONG

Department of Computer Science & Engineering

REPORT

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Mapreduce And Pig
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1. Problem Statement

Analyze the LinkedIn Jobs & Skills dataset (1.3M+ job postings, 2024) using Hadoop (MapReduce) and Apache Pig based on different column fields to provide comprehensive insights into hiring trends, in-demand skills, and geographic job distribution.

2. Dataset Description

Dataset: <https://www.kaggle.com/datasets/asaniezka/1-3m-linkedin-jobs-and-skills-2024/data>

The dataset consists of three CSV files:

linkedin_job_postings.csv

Column	Description
job_link	Unique URL identifier of the job posting
last_processed_time	Timestamp of last data processing
got_summary / got_ner	Flags indicating whether summary/NER extraction was performed
is_being_worked	Processing status flag
job_title	Title of the job posting
company	Name of the hiring company
job_location	Location of the job
first_seen	Date the job posting first appeared
search_city / search_country	City/Country used for the search query
search_position	Search keyword/position used
job_level	Seniority level (Entry, Mid, Senior, etc.)
job_type	Employment type (Full-time, Part-time, etc.)

job_skills.csv

Column	Description
job_link	Unique URL identifier matching the postings file
job_skills	Comma-separated list of skills required for the job

job_summary.csv

Column	Description
job_link	Unique URL identifier matching the postings file
job_summary	Full text description/summary of the job posting

Note: Due to the large size of job_summary.csv (~5 GB), this project's analysis focuses on linkedin_job_postings.csv and job_skills.csv.

3. Project Repository

GitHub Repository: <https://github.com/naimuddin1190/LinkedIn-Jobs-Skills>

All source code (MapReduce Java programs and Pig scripts) used in this project is available in the repository above.

4. Summary of Data Analysis

The following MapReduce programs and Pig scripts were implemented to analyze the dataset:

MapReduce Analysis:

- Total Jobs by Company — count of job postings per company
- Total Jobs by Country — count of job postings per country (search_country)
- Top Skills in Demand — frequency count of each skill across all job postings

Pig Analysis:

- Top 10 Companies by number of job postings
- Top 10 most frequently posted Job Titles
- Job count grouped by Job Level
- Top 20 Skills in demand
- Job count grouped by Country

Environment Setup:

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>java -version
openjdk version "1.8.0_504"
OpenJDK Runtime Environment (Temurin)(build 1.8.0_504-b01)
OpenJDK 64-Bit Server VM (Temurin)(build 25.504-b01, mixed mode)

C:\Windows\System32>hadoop version
Hadoop 3.2.4
Source code repository Unknown -r 7e5d9983b388e372fe640f21f048f2f2ae6e9eba
Compiled by ubuntu on 2022-07-12T11:58Z
Compiled with protoc 2.5.0
From source with checksum ee031c16fe785bbb35252c749418712
This command was run using /C:/hadoop/share/hadoop/common/hadoop-common-3.2.4.jar

C:\Windows\System32>pig -version
Apache Pig version 0.18.0 (r1926484)
compiled Jun 16 2025, 10:42:41

C:\Windows\System32>
```

5. Map Reduce Analysis

Each MapReduce program consists of three classes:

- Driver Class — configures the job (Mapper, Reducer, input/output paths)
- Mapper Class — reads each CSV record, extracts the relevant field, and emits a key-value pair
- Reducer Class — aggregates values for each key and writes the final count

5.1 Total Jobs by Company

Returns the total number of job postings for each company (Company $\times$ Job Count)

Driver Class

```
package jobsbycompany;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class DriverClass {
    public static void main(String[] args) throws Exception {
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "JobsByCompany");
        job.setJarByClass(DriverClass.class);

        job.setMapperClass(MapperClass.class);
        job.setReducerClass(ReducerClass.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Class

```
package jobsbycompany;

import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class MapperClass extends Mapper<LongWritable, Text, Text, IntWritable> {

    private Text company = new Text();
    private final static IntWritable one = new IntWritable(1);

    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        // Skip header row
        if (line.startsWith("job_link") || line.contains("search_country")) {
            return;
        }

        // Quote-aware split: does not treat comma inside quotes as a field separator
        String[] fields = line.split("(?=(?:[^\"]*" + "[^"]*" + "\\")*"[^"]*" + "$)", -1);

        if (fields.length > 6) {
            String companyName = fields[6].trim();

            // Remove surrounding quote characters if present
            if (companyName.startsWith("\"") && companyName.endsWith("\"") &&
companyName.length() > 1) {
                companyName = companyName.substring(1, companyName.length() - 1);
            }
            companyName = companyName.replace("\"\"", "").trim();

            if (!companyName.isEmpty()) {
                company.set(companyName);
                context.write(company, one);
            }
        }
    }
}
```

Reducer Class

```
package jobsbycompany;

import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class ReducerClass extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {

        int sum = 0;
        for (IntWritable val : values) {
            sum += val.get();
        }
        context.write(key, new IntWritable(sum));
    }
}
```

Execution

```
F:\BDALFINAL\LinkedIn Jobs & Skills>hadoop fs -cat /LinkedInProject/output/jobsbycompany_full/part-r-00000 | more
#twiceasnice Recruiting 179
$100B+ Alternative Asset Manager 1
$300B AUM Real Estate Firm 1
$300B+ Alternative Investment Firm 1
$50 Million Manufacturing Company 1
$500 Million Manufacturing/Distribution company 1
& Other Stories 1
&ahead 1
'Alohilani Resort Waikiki Beach 9
'SMITHS' of Smithfield 1
(ATI) Advanced Technology Institute 1
(GWI) GOODWILL INTERNATIONAL 12
(MALIN+GOETZ) 1
(Mindlance) 3
(NIDA) National Institute of Dramatic Art 3
(SLS) Scientific Laboratory Supplies Ltd. 1
(USTA) United States Tennis Association 7
(WIS) Wellbore Integrity Solutions 5
+61 1
+MEDRITE Urgent Care 2
. 21
.. 1
1 Hotel Nashville 2
1 Hotels 3
1 POINT SYSTEM LLC 1
1 Priority Environmental Services LLC. 2
1 Resource Group Staffing 50
1 Source 3
1 Tech Staffing Ltd 3
1+1 Cares 1
1-800-FLOWERS.COM, INC. 39
1-800-PACK-RAT 1
1-800-Packouts of North Detroit 1
1-800-STRIPER™ of NW Pittsburgh 1
1-800-STRIPER™ of Queens 1
1-800Accountant 20
1/ST Technology - AmTote International and Xpressbet 1
10 B Street Sports 2
10 Design 1
10 Fitness 2
10 Percent Recruiting Ltd. 9
100 Percent 1
100% Chiropractic Franchise 5
1000heads 9
100x Imaging Inc. 15
1031 CORP. 1
10Beauty 2
10Pearls 6
10X Health System 5
10X Recruiting Partners 6
10bits 9
```

5.2 Total Jobs by Country

Returns the total number of job postings for each country (Country × Job Count)

Driver Class

```
package locationcount;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class DriverClass {
    public static void main(String[] args) throws Exception {
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "LocationCount");
        job.setJarByClass(DriverClass.class);

        job.setMapperClass(MapperClass.class);
        job.setReducerClass(ReducerClass.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Class

```
package locationcount;

import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class MapperClass extends Mapper<LongWritable, Text, Text, IntWritable> {

    private Text country = new Text();
    private final static IntWritable one = new IntWritable(1);

    @Override
```

```

public void map(LongWritable key, Text value, Context context)
    throws IOException, InterruptedException {

    String line = value.toString();

    // Skip header row
    if (line.startsWith("job_link") || line.contains("search_country")) {
        return;
    }

    // Quote-aware split
    String[] fields = line.split("(?=(?:[^\"]*"\"[^\"]*"\"[^\"]*$)", -1);

    if (fields.length > 10) {
        String countryName = fields[10].trim();

        if (countryName.startsWith("\"") && countryName.endsWith("\"") &&
countryName.length() > 1) {
            countryName = countryName.substring(1, countryName.length() - 1);
        }
        countryName = countryName.replace("\"\"", "").trim();

        if (!countryName.isEmpty()) {
            country.set(countryName);
            context.write(country, one);
        }
    }
}

```

Reducer Class

```
package locationcount;

import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class ReducerClass extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {

        int sum = 0;
        for (IntWritable val : values) {
            sum += val.get();
        }
        context.write(key, new IntWritable(sum));
    }
}
```

Execution

```
F:\BDALFINAL\LinkedIn Jobs & Skills>hadoop fs -cat /LinkedInProject/output/locationcount_full/part-r-00000 | more
Australia      29719
Canada  55971
United Kingdom 113391
United States  1149339
```

5.3 Top Skills in Demand

Returns the total occurrence count of each skill mentioned across job postings (Skill × Count)

Driver Class

```
package skillcount;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class DriverClass {
    public static void main(String[] args) throws Exception {
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "SkillCount");
        job.setJarByClass(DriverClass.class);

        job.setMapperClass(MapperClass.class);
        job.setReducerClass(ReducerClass.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Class

```
package skillcount;

import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class MapperClass extends Mapper<LongWritable, Text, Text, IntWritable> {

    private Text skill = new Text();
    private final static IntWritable one = new IntWritable(1);

    @Override
```

```

public void map(LongWritable key, Text value, Context context)
    throws IOException, InterruptedException {

    String line = value.toString();

    // Quote-aware split: keeps the quoted skills list as one field
    String[] fields = line.split("(?=(?:[^\"]*"\"[^\"]*"\"[^\"]*"*)\"[^\"]*$)", -1);

    if (fields.length < 2) return;

    String skillsField = fields[1].trim();

    if (skillsField.startsWith("\"") && skillsField.endsWith("\"") && skillsField.length() > 1)
    {
        skillsField = skillsField.substring(1, skillsField.length() - 1);
    }
    skillsField = skillsField.replace("\"\"", "");

    String[] skills = skillsField.split(",");

    for (String s : skills) {
        String cleaned = s.trim();
        if (!cleaned.isEmpty()) {
            skill.set(cleaned);
            context.write(skill, one);
        }
    }
}

```

Reducer Class

```

package skillcount;

import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class ReducerClass extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {

        int sum = 0;
        for (IntWritable val : values) {
            sum += val.get();
        }
        context.write(key, new IntWritable(sum));
    }
}

```

Execution

Administrator: Command Prompt

```
F:\BDALFINAL\LinkedIn Jobs & Skills>hadoop fs -cat /LinkedInProject/output/skillcount_full/part-r-00000 | more
"20 Tools of Process Control" 1
"2nd Approval" Opportunities 1
"6 days on and 2 days off guaranteed" schedule 1
"A school" or "C school" 1
"A" Grade Leapfrog Group 1
"A" Grade from Leapfrog Group 2
"A" Grade from the Leapfrog Group for Quality and Safety 1
"A" Hospital Safety Grade 1
"A" Register Maintenance 1
"A" type personality 1
"ALLon4" Surgeries 1
"Adventurer Mindset" 1
"As a Service" Sales Experience 1
"As built" records 1
"Asbuilt" drawings 1
"Asbuilts" development 1
"Asbuilts" maintenance 1
"Associate First" Culture 1
"At Will" state. 1
"B" Water Treatment / Distribution License 1
"Bad Actors" and "Opportunistic Maintenance" Lists 1
"Be FAIR" ideals 1
"Be FAIR" ideals: Friendly Accommodating Inclusive Respectful 1
"Bedside manner" 1
"Belief Day" 1
"Best Fit Every Time" policy 1
"Best Places to Live." 1
"Best practice" 1
"Big 4" Experience 1
"Big 4" or National Accounting Firm Experience 1
"Big 4" or national accounting firm 1
"Big Box" store management 1
"Big Four" Audit Experience 1
"Big Joe" operation 4
"Big Picture" Thinking 1
"Bigbox" retail experience 1
"Bigbox" retail management 7
"Bigbox" retail management experience 2
"Birthday Leave" 1
"Build to" charts 1
"By any means necessary" spirit 2
"C" 1
"C" Class Driver's License 1
"C" Class Drivers Licence 1
"C" Class Drivers License 1
"C" Private Investigator License 1
"C" level executives 1
"CGI Advantage" ERP Implementation Experience 1
"CTuit" system 1
"Caf|- " 1
"Can Do Attitude" 2
"Can Do" Approach 2
"Can Do" Attitude 17
"Can Do" attitude 7
"Can do" attitude 27
"Can do" attitude and "will do" work ethic 1
"CanDo" Attitude 7
"Cando attitude" 2
"Cando" Attitude 1
"Cando" attitude 13
"Cando" demeanor 1
"Cando" openminded attitude 1
```

6. Pig Analysis

All Pig scripts were executed in local mode using the command: `pig -x local <script>.pig`

6.1 top10companies.pig

Returns the Top 10 companies by number of job postings

Script

```
postings = LOAD 'linkedin_job_postings.csv' USING PigStorage(',') AS
(job_link:chararray, last_processed_time:chararray, got_summary:chararray,
got_ner:chararray, is_being_worked:chararray, job_title:chararray,
company:chararray, job_location:chararray, first_seen:chararray,
search_city:chararray, search_country:chararray, search_position:chararray,
job_level:chararray, job_type:chararray);

filtered = FILTER postings BY company IS NOT NULL AND company != 'company';
grouped = GROUP filtered BY company;
counted = FOREACH grouped GENERATE group AS company, COUNT(filtered.job_link)
AS total_jobs;
sorted = ORDER counted BY total_jobs DESC;
top10 = LIMIT sorted 10;

DUMP top10;
```

Execution

```
Administrator Command Prompt

Success!

Job Stats (time in seconds):
JobId Maps Reduces MaxMapTime MinMapTime AvgMapTime MedianMapTime MaxReduceTime MinReduceTime AvgReduceTime MedianReduceTime Alias Feature Outputs
job_local1006327200_0004 1 1 n/a n/a n/a n/a n/a n/a sorted file:/tmp/temp532655367/tmp696259901,
job_local320933493_0002 1 1 n/a n/a n/a n/a n/a n/a sorted SAMPLER
job_local357080138_0001 13 1 n/a n/a n/a n/a n/a n/a counted,filtered,grouped,postings GROUP_BY,COMBINER
job_local837696320_0003 1 1 n/a n/a n/a n/a n/a n/a sorted ORDER_BY,COMBINER

Input(s):
Successfully read 1348511 records from: "file:///F:/BDALFINAL/LinkedIn Jobs & Skills/linkedin_job_postings.csv"

Output(s):
Successfully stored 10 records in: "file:/tmp/temp532655367/tmp696259901"

Counters:
Total records written : 10
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local357080138_0001 -> job_local320933493_0002,
job_local320933493_0002 -> job_local837696320_0003,
job_local837696320_0003 -> job_local1006327200_0004,
job_local1006327200_0004

2026-09-14 23:55:00,628 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:55:00,629 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:55:00,630 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:55:00,634 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:55:00,636 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:55:00,637 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:55:00,639 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:55:00,645 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:55:00,646 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:55:00,649 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:55:00,650 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:55:00,651 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:55:00,653 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning ACCESSING_NON_EXISTENT_FIELD 663 time(s).
2026-09-14 23:55:00,653 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-14 23:55:00,656 [main] WARN org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initialized
2026-09-14 23:55:00,658 [main] WARN org.apache.hadoop.io.nativeio.NativeIO - NativeIO.getStat error (3): The system cannot find the path specified.
-- file path: tmp/temp532655367/tmp696259901/part-r-00000
2026-09-14 23:55:00,684 [main] WARN org.apache.hadoop.io.nativeio.NativeIO - NativeIO.getStat error (3): The system cannot find the path specified.
-- file path: tmp/temp532655367/tmp696259901/SUCCESS
2026-09-14 23:55:00,710 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-14 23:55:00,710 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
("Health eCareers",34216)
("Jobs for Humanity",23545)
("Dollar General",14814)
("TravelNurseSource",14349)
("Jobot",8553)
("Gotham Enterprises Ltd",8417)
("McDonald's",7918)
("ClearanceJobs",7692)
("VolunteerMatch",6407)
("PracticleLink",6019)
2026-09-14 23:55:00,756 [main] INFO org.apache.pig.Main - Pig script completed in 16 seconds and 261 milliseconds (16261 ms)
```

6.2 top10titles.pig

Returns the Top 10 most frequently posted job titles

Script

```
postings = LOAD 'linkedin_job_postings.csv' USING PigStorage(',') AS
(job_link:chararray, last_processed_time:chararray, got_summary:chararray,
got_ner:chararray, is_being_worked:chararray, job_title:chararray,
company:chararray, job_location:chararray, first_seen:chararray,
search_city:chararray, search_country:chararray, search_position:chararray,
job_level:chararray, job_type:chararray);

filtered = FILTER postings BY job_title IS NOT NULL AND job_title != 'job_title';
grouped = GROUP filtered BY job_title;
counted = FOREACH grouped GENERATE group AS title, COUNT(filtered.job_link) AS
total;
sorted = ORDER counted BY total DESC;
top10 = LIMIT sorted 10;

DUMP top10;
```

Execution

```
Administrator: Command Prompt
Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature Outputs
job_local1497017758_0001  13  1  n/a  n/a  n/a  n/a  n/a  n/a  counted,filtered,grouped,postings  GROUP_BY_COMBINER
job_local1669615331_0003  1  1  n/a  n/a  n/a  n/a  n/a  n/a  sorted  ORDER_BY_COMBINER
job_local1351558525_0002  1  1  n/a  n/a  n/a  n/a  n/a  n/a  sorted  SAMPLER
job_local1959754818_0004  1  1  n/a  n/a  n/a  n/a  n/a  n/a  sorted  file:/tmp/temp1828125270/tmp1717516885,

Input(s):
Successfully read 1348511 records from: "file:///F:/BDALFINAL/LinkedIn Jobs & Skills/linkedin_job_postings.csv"

Output(s):
Successfully stored 10 records in: "file:/tmp/temp1828125270/tmp1717516885"

Counters:
Total records written : 10
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1497017758_0001 -> job_local1351558525_0002,
job_local1351558525_0002 -> job_local1669615331_0003,
job_local1669615331_0003 -> job_local1959754818_0004,
job_local1959754818_0004

2026-09-14 23:56:53,041 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:56:53,042 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:56:53,043 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:56:53,049 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:56:53,051 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:56:53,052 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:56:53,055 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:56:53,061 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:56:53,062 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:56:53,065 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:56:53,066 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:56:53,068 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:56:53,070 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-14 23:56:53,070 [main] WARN org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initialized
2026-09-14 23:56:53,073 [main] WARN org.apache.hadoop.io.nativeio.NativeIO - NativeIO.getStat error (3): The system cannot find the path specified.
-- file path: tmp/temp1828125270/tmp1717516885/part-r-00000
2026-09-14 23:56:53,106 [main] WARN org.apache.hadoop.io.nativeio.NativeIO - NativeIO.getStat error (3): The system cannot find the path specified.
-- file path: tmp/temp1828125270/tmp1717516885/_SUCCESS
2026-09-14 23:56:53,133 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-14 23:56:53,133 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
("LEAD SALES ASSOCIATE-PT",7325)
("Manager",6573)
("Shift Manager",5818)
("First Year Tax Professional",5356)
("Assistant Manager",5346)
("Customer Service Representative",5203)
("LEAD SALES ASSOCIATE-PT",4924)
("Store Manager",4791)
("CUSTOMER SERVICE REPRESENTATIVE",4218)
("Registered Nurse",4190)
2026-09-14 23:56:53,175 [main] INFO org.apache.pig.Main - Pig script completed in 19 seconds and 23 milliseconds (19023 ms)
```

6.3 jobstylelevel.pig

Returns the total number of job postings grouped by job level (Entry, Mid, Senior, etc.)

Script

```
postings = LOAD 'linkedin_job_postings.csv' USING PigStorage(',') AS
(job_link:chararray, last_processed_time:chararray, got_summary:chararray,
got_ner:chararray, is_being_worked:chararray, job_title:chararray,
company:chararray, job_location:chararray, first_seen:chararray,
search_city:chararray, search_country:chararray, search_position:chararray,
job_level:chararray, job_type:chararray);

filtered = FILTER postings BY job_level IS NOT NULL AND job_level != 'job_level';
grouped = GROUP filtered BY job_level;
counted = FOREACH grouped GENERATE group AS level, COUNT(filtered.job_link) AS
total;
sorted = ORDER counted BY total DESC;

DUMP sorted;
```

Execution

```
Administrator: Command Prompt
Input(s):
Successfully read 1348511 records from: "file:///F:/BDALFINAL/LinkedIn Jobs & Skills/linkedin_job_postings.csv"

Output(s):
Successfully stored 3441 records in: "file:/tmp/temp-1946084356/tmp1073098765"

Counters:
Total records written : 3441
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1536429533_0001 -> job_local101830623_0002,
job_local101830623_0002 -> job_local1090698112_0003,
job_local1090698112_0003

2026-09-14 23:58:10,688 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:58:10,691 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:58:10,692 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:58:10,698 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:58:10,701 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:58:10,702 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:58:10,705 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:58:10,706 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:58:10,706 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-14 23:58:10,708 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning ACCESSING_NON_EXISTENT_FIELD 663 time(s).
2026-09-14 23:58:10,708 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-14 23:58:10,709 [main] WARN org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initialized
2026-09-14 23:58:10,713 [main] WARN org.apache.hadoop.io.nativeio.NativeIO - NativeIO.getStat error (3): The system cannot find the path specified.
-- file path: tmp/temp-1946084356/tmp1073098765/part-r-00000
2026-09-14 23:58:10,749 [main] WARN org.apache.hadoop.io.nativeio.NativeIO - NativeIO.getStat error (3): The system cannot find the path specified.
-- file path: tmp/temp-1946084356/tmp1073098765/ SUCCESS
2026-09-14 23:58:10,776 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-14 23:58:10,776 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
("United States",197347)
("United Kingdom",103027)
("Canada",42999)
("Australia",25716)
("Mid senior",12095)
("Account Executive",11552)
("Christian Science Nurse",10448)
("Change Person",8430)
("Circulation-Sales Representative",8269)
("Consultant Education",7368)
("Extra",6671)
("Car Inspector",6617)
("Accountant",6509)
("Check Cashier",6245)
("Bakery Supervisor",6219)
("Chef",6106)
("Dermatologist",5863)
("Anesthesiologist",5810)
("Agricultural-Research Engineer",5683)
("Nurse Practitioner",5652)
("Nurse-Midwife",5622)
("Manager Customer Service",5577)
("Assembler",5513)
("Manager Benefits",5473)
("Counselor",5353)
("Nurse Infection Control",5340)
```

6.4 top20skills.pig

Returns the Top 20 most in-demand skills across all job postings

Script

```
skills = LOAD 'job_skills.csv' USING PigStorage(',') AS
  (job_link:chararray, job_skills:chararray);

filtered = FILTER skills BY job_skills IS NOT NULL AND job_link != 'job_link';
tokenized = FOREACH filtered GENERATE FLATTEN(TOKENIZE(job_skills, ',')) AS skill;
cleaned = FOREACH tokenized GENERATE TRIM(skill) AS skill;
grouped = GROUP cleaned BY skill;
counted = FOREACH grouped GENERATE group AS skill, COUNT(cleaned) AS total;
sorted = ORDER counted BY total DESC;
top20 = LIMIT sorted 20;

DUMP top20;
```

Execution

```
Administrator: Command Prompt
Input(s):
Successfully read 1296382 records from: "file:///F:/BDALFINAL/LinkedIn Jobs & Skills/job_skills.csv"

Output(s):
Successfully stored 20 records in: "file:/tmp/temp-2053516183/tmp-1596870484"

Counters:
Total records written : 20
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local238476572_0001 -> job_local697241359_0002,
job_local697241359_0002 -> job_local57665919_0003,
job_local57665919_0003 -> job_local1187303621_0004,
job_local1187303621_0004

2026-09-15 00:00:13,854 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:00:13,855 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:00:13,857 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:00:13,863 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:00:13,868 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:00:13,869 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:00:13,872 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:00:13,872 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:00:13,873 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:00:13,874 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:00:13,874 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:00:13,875 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:00:13,876 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-15 00:00:13,878 [main] WARN org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initialized
2026-09-15 00:00:13,880 [main] WARN org.apache.hadoop.io.nativeio.NativeIO - NativeIO.getStat error (3): The system cannot find the path specified.
-- file path: tmp/temp-2053516183/tmp-1596870484/part-r-00000
2026-09-15 00:00:13,906 [main] WARN org.apache.hadoop.io.nativeio.NativeIO - NativeIO.getStat error (3): The system cannot find the path specified.
-- file path: tmp/temp-2053516183/tmp-1596870484/_SUCCESS
2026-09-15 00:00:13,931 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-15 00:00:13,931 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
("Customer service,62063)
("Nursing,60275)
("Sales,27250)
("Leadership,21388)
("Customer Service,16950)
("Registered Nurse,16895)
("Project Management,13868)
("Accounting,12884)
("Communication,11854)
("Communication skills,11518)
("Civil Engineering,4852)
("Nurse Practitioner,4781)
("Microsoft Office,4721)
("Electrical Engineering,4598)
("Retail experience,4596)
("Physician,4534)
("Registered Nurse (RN),4334)
("CT Technologist,4089)
("Management,4051)
("Dentistry,4022)
2026-09-15 00:00:13,970 [main] INFO org.apache.pig.Main - Pig script completed in 16 seconds and 396 milliseconds (16396 ms)
```

6.5 jobsbycountry.pig

Returns the total number of job postings grouped by country

Script

```
postings = LOAD 'linkedin_job_postings.csv' USING PigStorage(',') AS
(job_link:chararray, last_processed_time:chararray, got_summary:chararray,
got_ner:chararray, is_being_worked:chararray, job_title:chararray,
company:chararray, job_location:chararray, first_seen:chararray,
search_city:chararray, search_country:chararray, search_position:chararray,
job_level:chararray, job_type:chararray);

filtered = FILTER postings BY search_country IS NOT NULL AND search_country !=
'search_country';
grouped = GROUP filtered BY search_country;
counted = FOREACH grouped GENERATE group AS country, COUNT(filtered.job_link) AS
total;
sorted = ORDER counted BY total DESC;

DUMP sorted;
```

Execution

```
Select Administrator: Command Prompt

Input(s):
Successfully read 1348511 records from: "file:///F:/BDALFINAL/LinkedIn Jobs & Skills/linkedin_job_postings.csv"

Output(s):
Successfully stored 3416 records in: "file:/tmp/temp-1210404724/tmp1718554921"

Counters:
Total records written : 3416
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1978283707_0001 -> job_local615307667_0002,
job_local615307667_0002 -> job_local886698426_0003,
job_local886698426_0003

2026-09-15 00:01:21,087 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:01:21,089 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:01:21,090 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:01:21,096 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:01:21,101 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:01:21,102 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:01:21,104 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:01:21,105 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:01:21,106 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-15 00:01:21,108 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning ACCESSING_NON_EXISTENT_FIELD 663 time(s).
2026-09-15 00:01:21,108 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-15 00:01:21,110 [main] WARN org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initialized
2026-09-15 00:01:21,112 [main] WARN org.apache.hadoop.io.nativeio.NativeIO - NativeIO.getStat error (3): The system cannot find the path specified.
-- file path: tmp/temp-1210404724/tmp1718554921/part-r-00000
2026-09-15 00:01:21,140 [main] WARN org.apache.hadoop.io.nativeio.NativeIO - NativeIO.getStat error (3): The system cannot find the path specified.
-- file path: tmp/temp-1210404724/tmp1718554921/_SUCCESS
2026-09-15 00:01:21,165 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-15 00:01:21,165 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
("2024-01-14",118924)
("2024-01-13",113529)
("2024-01-12",58348)
("2024-01-16",43810)
("2024-01-15",23483)
("United States",13434)
("2024-01-17",10994)
( Canada",8284)
("Garland",7896)
("North Carolina",7705)
("Baytown",7390)
("Austin",7027)
("South Carolina",6794)
( United Kingdom",6597)
("Indiana",6103)
("Alabama",5974)
("Atlanta",5841)
("Norristown",5762)
("Glendale",5702)
("Chicago",5668)
("Alexandria",5616)
("Avondale",5588)
("San Antonio",5505)
("Wisconsin",5462)
("Santa Clara",5395)
```

7. Conclusion

This project demonstrates the use of the Hadoop ecosystem to process and analyze a large-scale (1.3M+ record) real-world dataset. MapReduce programs were used to compute aggregate counts (jobs by company, jobs by country, and skill demand frequency) directly against raw CSV data, while Apache Pig was used to express similar and additional analyses (top companies, top job titles, job level distribution, top skills, and job distribution by country) using a higher-level scripting language with far less code. Together, these results provide insight into which companies are hiring most actively, which regions have the highest job volume, and which skills are most in demand in the current job market.

8. References

- [1] Kaggle Dataset — 1.3M LinkedIn Jobs & Skills (2024): <https://www.kaggle.com/datasets/asaniczka/1-3m-linkedin-jobs-and-skills-2024/data>
- [2] Project Source Code (GitHub): <https://github.com/naimuddin1190/LinkedIn-Jobs-Skills>
- [3] Apache Hadoop Documentation: <https://hadoop.apache.org/docs/stable/>
- [4] Apache Hadoop MapReduce Tutorial: <https://hadoop.apache.org/docs/stable/hadoop-mapreduce-client/hadoop-mapreduce-client-core/MapReduceTutorial.html>
- [5] Apache Pig Documentation: <https://pig.apache.org/docs/latest/>
- [6] Apache Pig Latin Reference Manual: <https://pig.apache.org/docs/latest/basic.html>